Exploring the Potential of a Centralised University Assessment Database

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***Abstract*— This project investigates whether a centralised, university-built repository could improve access to educational content for the university, and what could be built on top of it. Using assessment data as a test case, we developed a prototype that extracts information from assessment briefs, validates it with a Pydantic schema, and stores it in a MySQL database covering module information, learning outcomes, assessment requirements and marking rubrics, with a Retrieval-Augmented Generation (RAG) chatbot built on top.**

Keywords - centralised repository, assessment data, relational database, retrieval-augmented generation (RAG), structured data extraction, large language models (LLM).

I. Introduction

This project investigates whether and how a centralised, university-built repository could improve access to educational content for these three groups, and what could be built on top of it.

We use assessment data as a test case. We built a prototype that extracts information from assessment briefs in PDF and Word format, validates it against a Pydantic schema, and stores it in a MySQL relational database. The database holds module information, learning outcomes, assessment requirements and marking rubrics. Two applications were built on it: a chatbot that answers student questions and a tool that gives feedback on draft work against the marking criteria. Both use Retrieval-Augmented Generation (RAG), meaning the AI model answers from retrieved university documents and data instead of from its general knowledge. The prototype was tested on 19 briefs from different departments.

The report addresses four questions:

1. Can assessment information from separate documents be brought into one structured database?
2. Does centralisation give a clearer overview across modules, and can it reveal wider issues in assessment briefs?
3. What applications can be built on one shared repository, such as student and staff tools, educational analytics and AI?
4. How does building a system in-house compare with buying one in, in terms of cost and security?

The prototype provides evidence for the first three questions. The fourth, along with the use of analytics to identify where students and lecturers struggle, is discussed through analysis because the prototype cannot measure it.

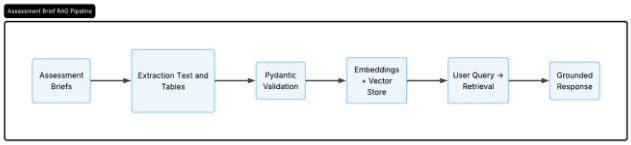
The report is written so that others at the university can reuse the work. It describes what was built, why each design choice was made, how it performed, and the problems we met. On 19 briefs, the prototype extracted majority of fields correctly, and the central database supported queries that separate documents could not, such as clashing deadlines and missing information. The main limitations are the small sample from one department, the quality of the source documents, and the use of an external AI service.

II. Problems and Aims

A. Problems

In universities assessment information is kept in separate documents for each department . Lecturers create briefs for each module and upload it to Moodle. This limits the access to information that the students, lecturers and administrative staff have. It makes it difficult to see deadlines, weightings, types of assessments and requirements across a programme, specifically on how they all relate. Students, lecturers and administrative staff are all affected by data silos which are defined as the hoarding of information within specific departments [2]. With 67% of global higher education institutions struggling with this it is clear to see that data that is isolated is common across departments; this hinders information sharing among departments and leads to decreased work efficiency and decision-making quality [1].

This also directly affects progressive learning in universities. Students find it difficult to get feedback and build relationships with lecturers, especially in cohorts with a high



volume, only receiving feedback after the submission of an assignment. This could be addressed via a tool created on top of a centralised assessment repository that gives draft feedback or answers questions. If a student wanted to see their upcoming deadlines, they would have to organise that themselves. A student also would have to relate the content of each brief and module together.

Lecturers have no insight into how students have understood the brief and how accessible or usable their assessment is. They also have no insight into which questions are asked most frequently about assessments which can reflect module teaching.

Administrators can't see whether lecturers are using the assigned templates for brief creation or not. Also, what the most common assessment types used by modules are. Analytics on how early or late students are submitting are also not known.

B. Aims

Looking at these issues we aim to investigate whether a centralised assessment repository will improve access to assessment information. We also will evaluate whether the

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data collected can be used to identify wider issues within modules, teaching or assessment briefs. On a wider scale implementation of a centralised database, it's important to ask what impact it will have.

Once the prototype we will investigate what kinds of tools could potentially be built on top. If a centralised repository will allow for effective integration across different departments and how will the university have its own university-built system to help with cost and security in comparison to importing from external companies. The objectives include:

- Build a prototype that stores assessment briefs in one structured database
- Evaluate how accurately it extracts the information.
- Build tools on top for students and staff.
- Discuss cost, security and wider use.

III. Approach and Prototype

The main aim of the prototype was a structured database of assessment information. Information is extracted from a set of assessment briefs, validated against a Pydantic schema, and stored in a MySQL relational database. We checked the extraction by querying the database in MySQL Workbench and comparing the results with the original briefs. A chatbot was then built on top as an example application.

A. Schema

We defined the Pydantic schema before extraction, to identify the key pieces of information in each brief. Starting from one brief and the standard template, we separated the information into discrete facts (module code, title, deadline, word count, weighting, submission format) and texts and lists (learning outcomes, task description, guidance, marking criteria)

The texts split further into long-form text and tables. The marking rubric, a grid of criteria and grade bands, needed its own nested class inside the marking criteria list.

Validators check values such as dates and word counts. Where a brief gives a deadline only as wording, such as "Week 10", both the parsed date and the original wording are stored. We tested the schema by checking whether any information in the brief had no field to go into and revised it until nothing was left out.

B. Extraction

Clean text was first extracted from each file, using PyMuPDF for PDFs and python-docx for Word documents. The text was then sent to Gemini, which fills in the schema.

We chose Gemini because the university already has a partnership with it. The model is told to use only information stated in the document and to return null otherwise, and Pydantic validates its output.

We tested the pipeline on one brief, checked it field by field, and then ran it on further briefs to see how well it generalised.

C. Database

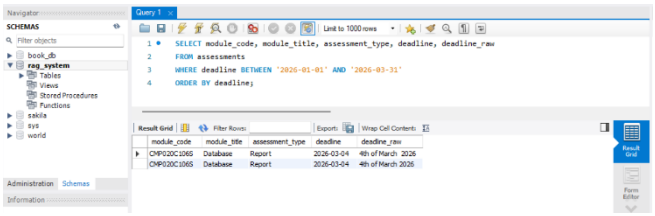
Validated records are stored in a MySQL relational database. Single-value facts sit in one assessments table. Lists (learning outcomes, checklist items, marking criteria and their grade bands) sit in linked tables, so each item can be counted and compared. Each record keeps its source file, so it can be traced back to the original document.

D. Search & Queries

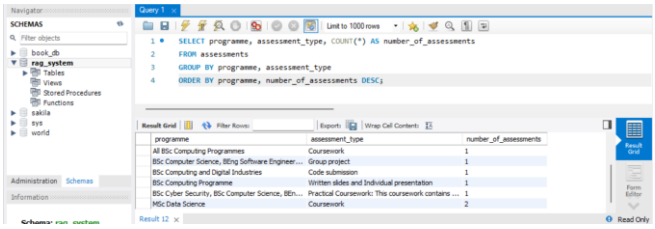
SQL queries answer cross-module questions that are hard to answer from separate documents. These include assessments within a time period, assessment types across a programme, modules with several assessments, weightings, missing information, and deadlines that are close together..

For example:

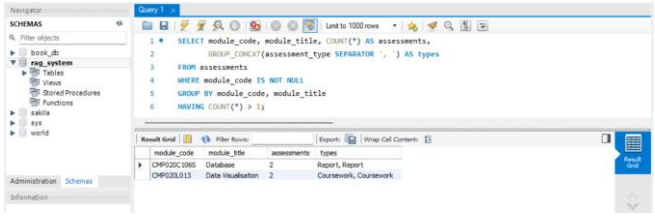
Assessments within a particular time



Assessment types across a programme



Modules with multiple assessments



E. Chatbot

The chatbot is a Streamlit app. When a student selects an assessment, facts, criteria and learning outcomes come directly from the database. The long text sections are searched with a local embedding model (all-MiniLM-L6-v2), restricted to that assessment.

Gemini then answers using only this material and says when the answer is not in the brief. We tested it with 5 questions covering facts, rules, meaning and information not in the brief, recording a pass or failure and the cause of each failure. Minimal questions were essential because of frequent 503 errors (temporarily unable to handle a request because of a server overload). We also compared the results with a baseline that sends the whole brief to the mode

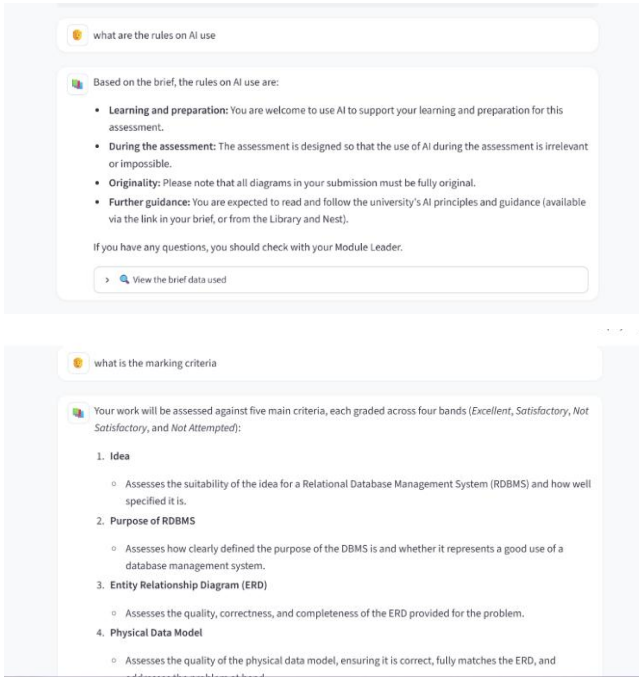
F. Feedback

For draft feedback, the system loads the selected assessment's criteria and grade bands from the database and gives them, with the student's draft, to the model as a checklist.

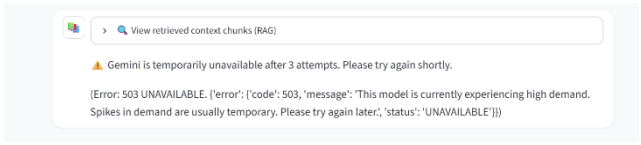
For each criterion, the model names the grade band the draft currently matches, with evidence from the draft, and says what would reach the next band. The word count is calculated

in code rather than by the model, and the tool gives feedback only, not a mark.

Pass example:



Failure example:



IV. Results

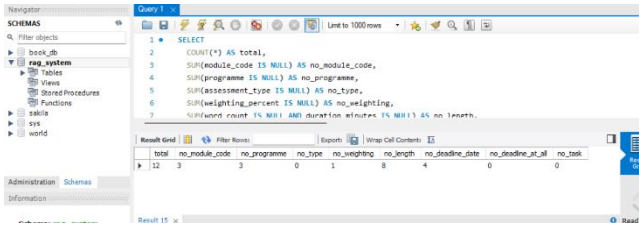
A. Ectraction Accuracy

We ran the pipeline on 19 assessment briefs (13 PDF, 6 Word) from multiple departments. Each stored record was compared field by field with the original document.

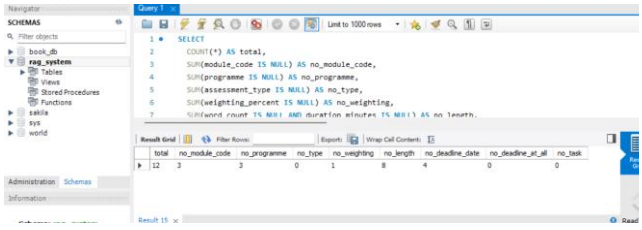
B. Cross-module queries

With 11 briefs in one database, SQL queries answered questions that separate documents could not answer directly.

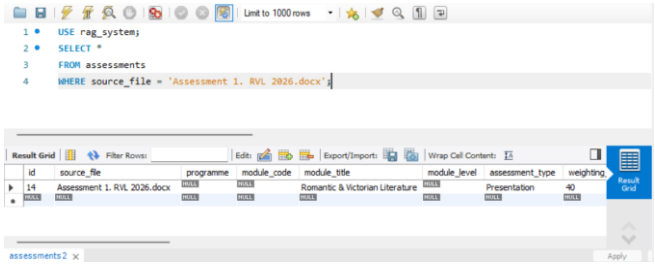
Deadlines that are close together



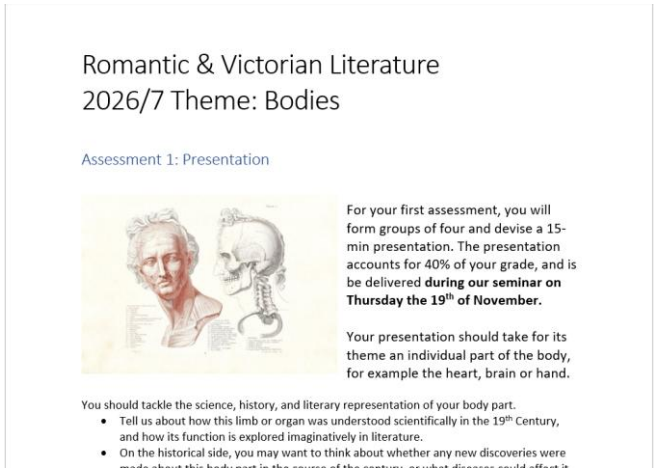
Missing assessment information



The missing-information queries also exposed gaps in the briefs themselves, such as:



, showing that a central database can reveal inconsistencies in how briefs are written. This brief is clearly missing the key identify features of a brief as shown below:



C. Chatbot

We wrote 5 test questions in four groups: facts, rules, meaning, and information not in the brief. We hoped to compare a baseline that sends the whole brief to the model with retrieval, which sends database facts plus the most relevant text sections. However long running time and 503 errors limited the ability to collect results

D. Costs & Security

Using a university-managed centralised repository and internally approved tools could offer benefits in terms of both cost and security. Rather than relying entirely on external services such as Gemini or ChatGPT, the university would have greater control over how assessment materials and student data are stored, accessed and managed. It could also reduce the risk of sensitive information being shared with third-party providers.

In terms of costs, developing and maintaining an internal system would require an initial investment in infrastructure, development and ongoing maintenance. However, it could reduce long-term reliance on external API subscriptions and usage-based pricing. A centralised repository could also reduce duplication by allowing different departments to access and reuse existing resources.

Although external tools may offer quicker implementation and access to advanced AI models, they introduce additional considerations around data protection, licensing, API costs and reliance on third-party services. An internally managed solution could provide greater institutional control and consistency, provided the university has the resources and expertise to maintain it.

V. Evaluation

A. What worked

Separating structured facts from free text let deadlines, weightings and criteria be answered exactly from the database, with search needed only for long descriptions. Validating against a schema made gaps visible instead of hiding them, and keeping each record's source file, with both the parsed date and the original wording, kept the data traceable. Caching retries and a fallback model made batch ingestion practical despite frequent service overload errors.

B. Limitations

- The sample was small and came from one department, so other layouts may need changes to the schema or prompts.
- Content that exists only as images was not captured.
- The evaluation was done by one person. It involved no students or lecturers.
- Brief text and student drafts were sent to an external AI service. A real deployment would need a university-approved or in-house model, and until then this weakens the security argument for building in-house.".

C. Research Questions

TABLE I. WHAT THE PROTOTYPE SHOWED

Question	***Evidence***	***Outcome***
Can assessment information be held in one structured database?	Section III – C	Supported for [department]; not tested elsewhere
Does centralisation give an overview across modules?	Section III -D	Supported
Can the database support tools for students and staff?	Section IV – C	Working prototype; not tested with users
Can the data reveal wider issues in briefs?	Section IV-B	Party: issues in briefs only
Can AI identify where students and lecturers struggle?	None	Not tested: needs question logs and performance data

VI. Conclusion

We investigated whether assessment information from separate documents could be centralised in one structured database, and what could be built on it. The database supported cross-module queries that scattered documents could not, and a chatbot and draft-feedback tool were built by reading from it instead of reprocessing documents, showing that one repository can serve several applications. The main limits are the server being overwhelmed with requests which slow down response time , the quality of the source briefs, and the use of an external AI service. Future work should extend the approach to more departments and document types, add module and performance data under proper governance to support analytics, test the tools with students and staff, and host the AI components within the university. Overall, the prototype suggests a centralised assessment database is feasible and useful as a foundation, provided limitations are addressed.

Acknowledgment

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References

[1] Y. Zhang, 'A Study of the Digital Transformation and Efficiency Improvement Path of University Administration', SHS Web of Conferences, vol. 213, p. 01001, 2025, doi: 10.1051/shsconf/202521301001.

[2] E. A. Onyame, S. Annim, A.-G. Sayibu, D. S. Y. Amedahe, N. Borketey-Coffie, and H. N. Manu, 'Bridging Knowledge Silos at the University of Cape Coast: The Role of Cross-Departmental Knowledge Models in Enhancing Knowledge Sharing and Institutional Memory', Intelligent Information Management, vol. 17, no. 04, pp. 117–139, 2025, doi: 10.4236/iim.2025.174007.